

Online Supplement

Source-Scored Structural Initial Designs for Chance-Constrained Optimization of Ordered Policy Profiles

Authors' names are not included for peer review

EC.1. Finite Algorithm

Algorithm 1: Source-scored structural initial design

1. Generate the declared 64-profile outcome-free library on 128 regular midpoints using library seed 20261799, equal to family seed 20260808 plus 991.
2. For every source task and profile, compute the replicated objective mean and the Gaussian chance-margin statistic in (8).
3. Convert each statistic to deterministic within-task average-tie percentile ranks; average the ranks uniformly across source tasks.
4. Compute exact Voronoi-cell cosine coefficients $c_0, \dots, c_8$, divide c_k by $1 + .25k$, append all nine diagonal squares, and standardize the 18 coordinates over the library.
5. Append source safety and objective ranks. Select the first center by (11); select nine further centers by (12).
6. Linearly interpolate each selected profile to the declared target grid, clip to bounds, and deterministically round. Freeze these ten policies before the first target outcome.
7. Run any declared target backend. Freeze a distinct ordered shortlist before verification.
8. Test each shortlist candidate on its independent all-success stream. Deploy the first certificate; abstain if none certifies.

The stable tie order is safety rank, objective rank, profile identifier for the first center and finite-library index for farthest-first. Robust source feasibility is diagnostic only. There is no endpoint replacement, hidden scalarization, target-outcome repair, or target-oracle branch in a non-oracle arm.

EC.1.1. Supplementary comparison rules

Standardized DCT maximin computes the same 18-dimensional z_j as the source-scored atlas. It starts at the library medoid minimizing mean Euclidean distance to all 64 profiles and adds nine farthest-first centers in z ; the lowest library index breaks ties. The original unstandardized DCT maximin control applies these rules directly to ϕ .

The source-greedy portfolio uses the same replicated source records and within-task percentile ranks as the source-scored atlas. Define $\rho_{qj} = .5(r_{qj}^g + r_{qj}^f)$. Starting from $A_0 = \emptyset$, it selects

$$j_\ell \in \arg \min_{j \notin A_{\ell-1}} \frac{1}{m} \sum_{q=1}^m \min_{i \in A_{\ell-1} \cup \{j\}} \rho_{qi}, \quad A_\ell = A_{\ell-1} \cup \{j_\ell\}, \quad \ell = 1, \dots, 10. \quad (\text{EC.1})$$

Ties favor the candidate with the lower overall mean source score $m^{-1} \sum_q \rho_{qj}$, then the lower library index. This adapts greedy portfolio selection (Feurer et al. 2022) to the declared finite policy library using safety and objective ranks. Both supplementary designs use the original interpolation, target budget, and independent verifier. Their evaluations reuse the original target seeds after manuscript review.

EC.2. Additional Statistical Results

EC.2.1. Paired supplementary comparisons

Table EC.1 retains the paired certification counts for the two supplementary controls. Bootstrap resampling draws 20 paired task seeds within each of the eight fixed regimes, retaining equal regime weights, with 10,000 replicates. Each resolution is analyzed separately. Wins and losses refer to source-only and control-only true-feasible certificates; ties include both successes and both failures. These are post-review comparisons on previously used seeds. The standardized $d = 1000$ control reuses the first review diagnostic unchanged.

Table EC.1 Source-scored atlas minus supplementary controls in true-feasible certification. Differences and intervals are in percentage points.

d	Comparator	Difference (pp)	Stratified 95% CI	Wins	Losses	Ties
200	Standardized DCT maximin	+16.25	[11.88, 20.62]	28	2	130
200	Source-greedy portfolio	+11.25	[4.38, 18.12]	28	10	122
1000	Standardized DCT maximin	+15.62	[11.25, 20.00]	28	3	129
1000	Source-greedy portfolio	+11.25	[4.38, 18.12]	28	10	122
10000	Standardized DCT maximin	+15.62	[11.25, 20.00]	28	3	129
10000	Source-greedy portfolio	+8.12	[1.25, 15.00]	27	14	119

EC.2.2. Verifier thresholds

Table EC.2 gives exact power calculations. At $n = 80$, the Clopper–Pearson lower-bound rule equals the all-success rule. At $n = 160$, its preregistered sensitivity threshold permits two failures. The primary results retain the all-success protocol.

EC.2.3. Full sensitivity matrix

The full one-factor sensitivity matrix is reported in Table EC.3. Entries are ordered by the tested levels in the second column. Effective-rank levels are explicit overrides; all other rows include the frozen baseline level. The nonmonotone responses, the zero certificate rate at $\alpha = .01$, and the identical frequency-penalty results are retained. The latter follow from column standardization cancelling the positive frequency scaling, rather than from robustness to an effective metric parameter.

Recorded wall time is descriptive rather than a resource-normalized endpoint. On the mixed CPU cluster, the primary ten-point initial-design arms average 1.2–1.7 seconds per synthetic task, whereas 394-call

Table EC.2 Exact candidate-wise certificate power at budgets 80 and 160.

n	p	All-success power	CP threshold	Allowed failures	CP power
80	0.950	0.017	80	0	0.017
80	0.975	0.132	80	0	0.132
80	0.990	0.448	80	0	0.448
80	0.995	0.670	80	0	0.670
80	1.000	1.000	80	0	1.000
160	0.950	0.000	158	2	0.012
160	0.975	0.017	158	2	0.234
160	0.990	0.200	158	2	0.784
160	0.995	0.448	158	2	0.953
160	1.000	1.000	158	2	1.000

Table EC.3 One-factor sensitivity of the source-scored design at $d = 1000$. Each rate aggregates eight fixed regime strata and 20 independently seeded latent tasks per regime at one resolution.

Factor	Tested levels	Feasible (%)	Certified (%)
Effective rank	4/8/16/32	89.4/85.0/84.4/83.8	55.6/36.3/25.6/29.4
Chance level α	.01/.05/.1	95.0/95.0/95.0	0.0/46.2/29.4
Calibrated safe mass	.03/.08/.18	69.4/95.0/99.4	22.5/46.2/70.0
Initial-design size n_0	5/10/20	79.4/95.0/96.9	42.5/46.2/46.9
Source tasks	1/2/4	85.0/95.0/95.6	33.8/46.2/45.0
Profiles per source task	32/64/128	83.1/95.0/73.8	38.1/46.2/38.8
Source replications	1/3/5	92.5/95.0/92.5	45.0/46.2/45.6
Retained frequency K	4/8/16	81.9/95.0/95.6	43.1/46.2/48.1
Frequency penalty κ	0/.25/1	95.0/95.0/95.0	46.2/46.2/46.2
Safety-rank weight	.25/1/4	95.0/95.0/95.0	46.9/46.2/46.2
Objective-rank weight	.25/1/4	95.0/95.0/95.0	46.9/46.2/46.2
First-center safety weight	.25/.5/.75	62.5/95.0/95.0	32.5/46.2/45.0

source-free controls average about 59 seconds and the 394-call functional-SCBO arm has a median of about 677 seconds. The synthetic simulator is analytic and cheap, so these figures measure implementation overhead rather than the time of a physical simulator; simulation-call accounting remains the portable comparison.

EC.2.4. Task-seed strata audit

The registered generator includes a deterministic `task_seed mod 5` cosine component. Table EC.4 reports its realized categories rather than silently treating the finite sample as a balanced IID mixture. The analysis uses fixed regime strata and applies concentration separately at each resolution.

Table EC.4 Realized latent-category counts among the 160 independent task seeds. The same seeds are crossed with all three grid resolutions.

Regime	Category 0	1	2	3	4	Total
Aligned low frequency	3	8	3	1	5	20
Growing rank	4	7	6	3	0	20
Frequency shift	8	2	3	2	5	20
Permutation	11	0	3	5	1	20
Irregular grid	4	4	6	4	2	20
Piecewise smooth	5	4	6	4	1	20
Sparse high frequency	3	1	7	4	5	20
Misspecified target	3	6	6	2	3	20
Overall	41	32	40	25	22	160

Table EC.5 Native transfer pipelines at $N = 13$ and $n_0 = 10$ with the same 384-call source archive and historical v69 terminal protocol. Candidate budgets are (80, 128, 128), with an 8-call incumbent objective comparison. Each method retains its own initialization and online adaptation.

Native method	FactorShock	Inventory	Queue	Total	False
Safe F-PACOH	0/20	0/20	0/20	0/60	0
RGPE	14/20	16/20	0/20	30/60	0
Stacked GP	0/20	13/20	0/20	13/60	0
MTGP	1/20	0/20	0/20	1/60	0
FSBO	6/20	2/20	0/20	8/60	0
HyperBO	0/20	0/20	0/20	0/60	0
MetaBO	19/20	9/20	0/20	28/60	0
MALIBO	12/20	2/20	0/20	14/60	0

EC.2.5. Native end-to-end transfer

No method certifies a Queue task. These 60-task totals comprise 20 algorithmic seeds per fixed family and measure repeatability, not generalization to a population of domains. The v69 protocol is common to these eight pipelines and differs from the primary (80, 80, 80) verifier.

Table EC.6 Postdecision Energy temporal stability. Counts are descriptive and do not replace the declared empirical-window certificate.

Design	Certified	Block	Nonoverlap	Joint
generic dct maximin	54	49	54	49
natural constant grid	52	44	52	44
random low frequency	56	51	56	51
raw sobol	43	36	43	36
source atlas	57	52	57	52
source greedy portfolio	59	55	59	55
standardized generic dct maximin	49	42	49	42
target only dct space scbo	51	45	51	45

EC.2.6. Energy temporal audit

V5 preserves all 421 certificates under physically nonoverlapping windows and 374 under both the chronological-block and joint criteria. Shared calendars and only five regions preclude an iid market-population interpretation.

EC.3. Mechanism Attribution and New-Family Confirmation

Table EC.7 Diagnostic first-center and distance interventions on the original 160 tasks at each grid resolution. Medoid first centers are fixed across metrics; raw L2 uses the same public profiles and Voronoi quadrature.

Resolution: design	Coverage	True certificates	Loss
200: Medoid + augmented	137/160	56/160	0.216
200: Source-scored atlas	154/160	82/160	0.105
200: Source-best + raw L2	153/160	79/160	0.112
200: Source-best + structural	154/160	83/160	0.105
200: Medoid + structural	137/160	56/160	0.216
1000: Medoid + augmented	135/160	49/160	0.254
1000: Source-scored atlas	152/160	74/160	0.143
1000: Source-best + raw L2	133/160	68/160	0.250
1000: Source-best + structural	152/160	75/160	0.143
1000: Medoid + structural	135/160	49/160	0.254
10000: Medoid + augmented	117/160	46/160	0.352
10000: Source-scored atlas	134/160	71/160	0.240
10000: Source-best + raw L2	133/160	68/160	0.247
10000: Source-best + structural	134/160	72/160	0.240
10000: Medoid + structural	117/160	46/160	0.352

Table EC.8 Prospective results by source-family and public-library seed. Each row contains 80 tasks at $d = 1000$.

Family: design	Coverage	True certificates	Loss
20260930: Source-scored atlas	57/80	28/80	0.341
20260930: Source-greedy portfolio	70/80	22/80	0.168
20260930: Medoid + structural	47/80	7/80	0.459
20261031: Source-scored atlas	50/80	14/80	0.448
20261031: Source-greedy portfolio	71/80	9/80	0.132
20261031: Medoid + structural	33/80	6/80	0.677
20261132: Source-scored atlas	48/80	14/80	0.452
20261132: Source-greedy portfolio	71/80	18/80	0.169
20261132: Medoid + structural	58/80	11/80	0.407

Table EC.9 Prospective results by fixed mechanism, pooling ten tasks from each of the three new families. Each row contains 30 tasks.

Mechanism: design	Coverage	True certificates
aligned low frequency: Source-scored atlas	3/30	0/30
aligned low frequency: Source-greedy portfolio	30/30	2/30
aligned low frequency: Medoid + structural	17/30	0/30
growing effective rank: Source-scored atlas	26/30	8/30
growing effective rank: Source-greedy portfolio	30/30	9/30
growing effective rank: Medoid + structural	20/30	1/30
frequency support shift: Source-scored atlas	24/30	9/30
frequency support shift: Source-greedy portfolio	12/30	1/30
frequency support shift: Medoid + structural	25/30	8/30
coordinate permutation: Source-scored atlas	5/30	0/30
coordinate permutation: Source-greedy portfolio	30/30	2/30
coordinate permutation: Medoid + structural	19/30	4/30
irregular grid: Source-scored atlas	30/30	22/30
irregular grid: Source-greedy portfolio	30/30	19/30
irregular grid: Medoid + structural	11/30	1/30
piecewise smooth: Source-scored atlas	28/30	8/30
piecewise smooth: Source-greedy portfolio	30/30	8/30
piecewise smooth: Medoid + structural	20/30	8/30
sparse high frequency: Source-scored atlas	20/30	7/30
sparse high frequency: Source-greedy portfolio	30/30	6/30
sparse high frequency: Medoid + structural	6/30	0/30
misspecified target: Source-scored atlas	19/30	2/30
misspecified target: Source-greedy portfolio	20/30	2/30
misspecified target: Medoid + structural	20/30	2/30

Across the 240 tasks, source scoring consumes 60736 calls and standardized maximin 56960. Their calls per true-feasible certificate are 1084.6 and 2373.3; the portfolio consumes 63936 calls, or 1304.8 per true certificate. This is observed archive reuse rather than a hypothetical deployment count.

Table EC.10 Predefined library-size sensitivities on the first new family. Each row contains 80 paired tasks; support composition changes with size.

Library size: design	Coverage	True certificates	Loss
32: Source-scored atlas	56/80	28/80	0.354
32: Medoid + structural	57/80	16/80	0.415
128: Source-scored atlas	51/80	18/80	0.393
128: Medoid + structural	49/80	7/80	0.528

Table EC.11 Observed reuse cost in prospective synthetic confirmation. Each source design pays once for each of 24 archives serving ten tasks.

Design	Source calls	Cumulative calls	Calls/true certificate
Medoid + augmented	9216	66176	2757.3
Source-scored atlas	9216	60736	1084.6
Source-best + raw L2	9216	62016	1348.2
Source-best + structural	9216	60976	1129.2
Source-greedy portfolio	9216	63936	1304.8
Medoid + structural	0	56960	2373.3

Table EC.12 Corrected Energy V5 reuse across the actual 18 markets. Each of five geographic archives is charged once. Target cost and certificates are averaged over the five algorithm repeats within each market.

Design	Source calls	Cumulative calls	Expected certificates	Calls/certificate
source atlas	3840	6842.0	11.4	600.2
generic dct maximin	0	3258.0	10.8	301.7
standardized generic dct maximin	0	3242.0	9.8	330.8
source greedy portfolio	3840	6954.0	11.8	589.3

EC.4. Additional Source-Budget and Geometry Diagnostics

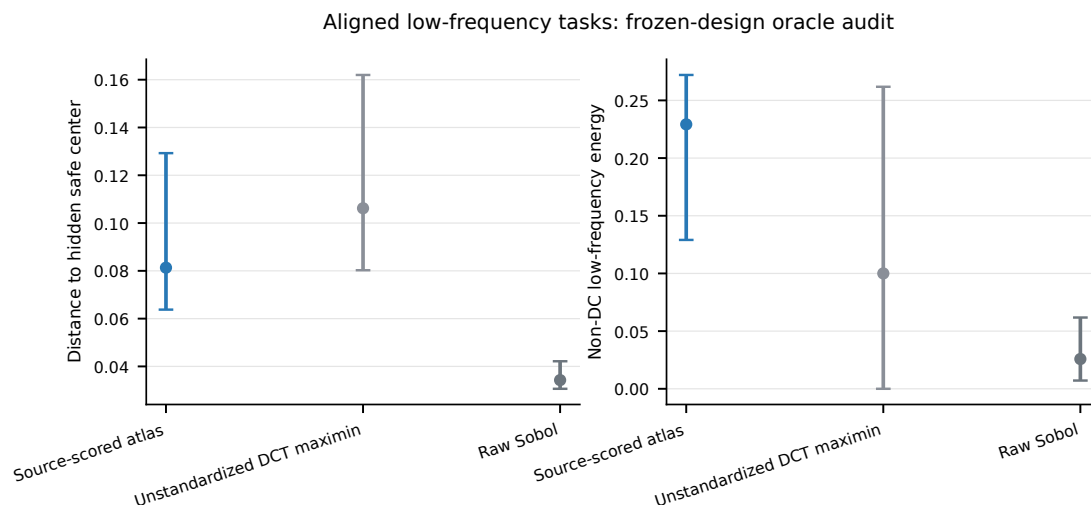


Figure EC.1 After-the-fact geometry diagnostic for the aligned regime. Points show medians and bars show 10th–90th percentiles over the original fixed designs. The hidden safe center was unavailable to every algorithm.

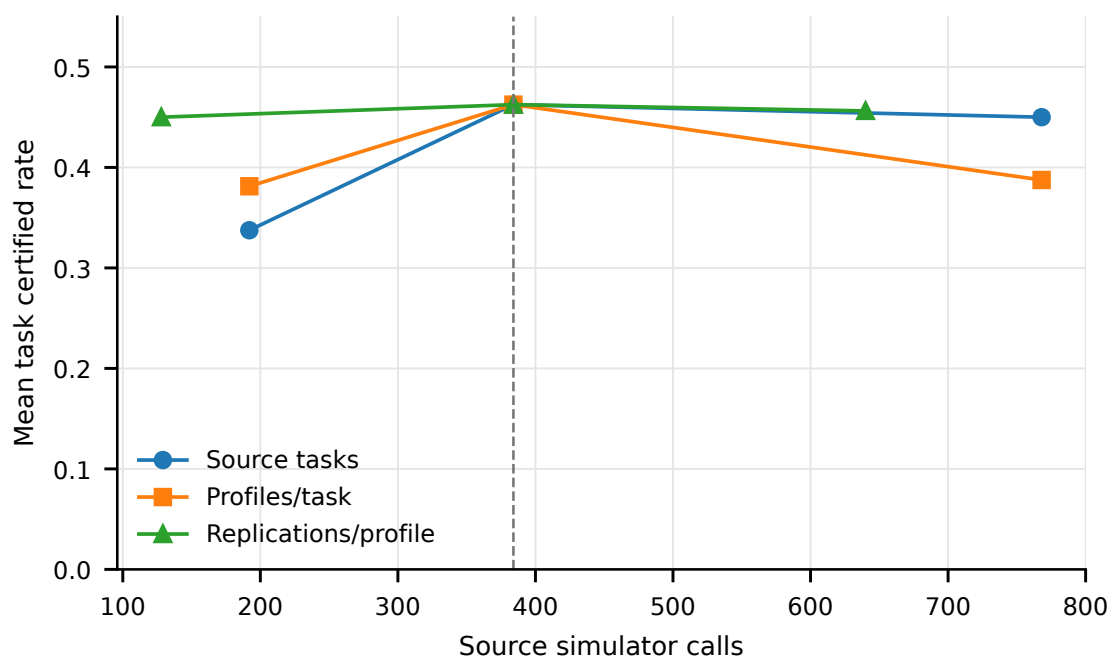


Figure EC.2 Source-budget sensitivity. More tasks, profiles, or replications do not monotonically improve the frozen design.

EC.5. Secondary Cumulative-Variance Diagnostic

The project originally studied cumulative heteroscedastic variance decomposition. Under matched policies, the factor-HVD model represents local and shared exposure and improves held-out variance-shape estimation. Table EC.13 gives the final matched diagnostic.

Table EC.13 Pooled variance versus cumulative factor-HVD. HVD improves calibration but not feasible recovery or verification cost.

Domain	Variance model	Feasible	Log-RMSE	Shape corr.	Mean verify
FactorShock	Pooled	20/20	1.158	0.000	159.2
FactorShock	Cumulative factor-HVD	20/20	0.219	0.919	158.4
Inventory	Pooled	20/20	0.577	0.000	182.4
Inventory	Cumulative factor-HVD	20/20	0.349	0.953	184.0
Queue	Pooled	19/20	0.501	0.000	190.4
Queue	Cumulative factor-HVD	19/20	0.254	0.995	197.6

Because both variance models recover the same feasible counts and HVD slightly increases verification cost in Inventory and Queue, HVD is not included in the paper's claimed optimization method. Its formal results remain in the proof archive as reusable mechanistic results.

EC.6. Human-Readable Proofs

Proof of Proposition 3. A normalized average-tie rank is $(J - 1)^{-1}$ times the sum, over all other profiles, of one for a lower score and one-half for an equal score. Under the simultaneous error event, a pair separated by more than $2r_q$ keeps its strict order. Only the $U_{qj}(r_q)$ unresolved pairs can change their contribution, each by at most one. This proves the rank bound. Weighted averages propagate these bounds to ϵ_g and ϵ_f .

Since $|\widehat{b}_j - b_j| \leq \epsilon_b$ and $\widehat{b}_{\widehat{j}} \leq \widehat{b}_{j^*}$ for a population minimizer, $b_{\widehat{j}} \leq b_{j^*} + 2\epsilon_b$. A population gap strictly larger than $2\epsilon_b$ forces the same unique minimizer. Structural coordinates are fixed, so each augmented coordinate moves by at most ϵ_{aug} and each pairwise distance by at most $2\epsilon_{\text{aug}}$. For a common selected set, taking the minimum over its members preserves this distance-error bound. A winner-to-runner-up gap larger than $4\epsilon_{\text{aug}}$ therefore preserves the next center. Induction from the common first center proves equality of the complete sequence. $\square$

EC.6.1. First-center uncertainty and target safety

For a simultaneous source-score bound $\max_j |\widehat{b}_j - b_j| \leq \epsilon_b$, define

$$C_b = \{j : \widehat{b}_j \leq \min_{\ell} \widehat{b}_{\ell} + 2\epsilon_b\}, \quad D_b = \max_{i,j \in C_b} \|\widehat{z}(h_i) - \widehat{z}(h_j)\|_2.$$

PROPOSITION EC.1 (First-center margin transfer). *Let j^* minimize b_j and $\widehat{j}$ minimize $\widehat{b}_j$. On the simultaneous score-error event, both belong to C_b . Suppose the target margin is L_q -Lipschitz in z_q^* , coordinate error is at most ϵ_η on this set, and $m_q(h_{j^*}) \leq -\gamma_q$. Then*

$$m_q(h_{\widehat{j}}) \leq -\gamma_q + L_q(D_b + 2\epsilon_\eta).$$

Thus $L_q(D_b + 2\epsilon_\eta) \leq \gamma_q$ suffices for a feasible first center. If the score-error event has probability at least $1 - \delta_s$ and the target premises hold throughout that event, the same lower bound applies to the probability of a feasible first center.

The Online Supplement gives the proof. Unlike complete-design stability, this result permits the first center to change within a geometrically small source-compatible set. It separates source estimation error from target transfer. Source data alone cannot identify whether the selected center is safer than a distinct outcome-free center: the same archive can accompany two targets assigning opposite safety margins to those policies.

Scope of the first-center condition. For a supplied valid source radius, C_b and D_b use only source scores and fixed coordinates. The target safety of j^* , L_q , and coordinate mismatch remain target premises. The source radii above depend on population scales and unresolved population-score pairs; substituting sample scales does not automatically yield a computable valid confidence radius. We do not use this analytic result to select policies or certify deployment in the experiments. A postdecision diagnostic on the 240 confirmation tasks evaluates the condition using exact source-score errors and a finite-library target Lipschitz constant. It

satisfies the sufficient condition on 0/240 tasks; the source-compatible sets contain 14–33 policies. Thus the condition clarifies the logical connection but does not explain the observed certification gain with a nonvacuous safety bound.

Proof of Proposition EC.1. Let k minimize $\widehat{b}$. On the source-score error event,

$$\widehat{b}_{j^\star} \leq b_{j^\star} + \epsilon_b \leq b_k + \epsilon_b \leq \widehat{b}_k + 2\epsilon_b.$$

Hence $j^\star \in C_b$; $\widehat{j} \in C_b$ by definition. The coordinate triangle inequality gives

$$\|z_q^\star(h_{\widehat{j}}) - z_q^\star(h_{j^\star})\|_2 \leq D_b + 2\epsilon_\eta.$$

Apply target Lipschitz continuity and the margin bound at $j^\star$. A nonpositive margin implies feasibility. If all target premises hold throughout the score-error event, that event is contained in the feasible-first-center event, proving the probability claim. $\square$

For first-center nonidentifiability, fix a source archive and two distinct policies a, c . On target 1 set deterministic margins $m_1(a) = -1$, $m_1(c) = 1$; on target 2 set $m_2(a) = 1$, $m_2(c) = -1$. All remaining library margins are 1 on both targets. Source outcomes, coordinates and source-only selections are unchanged, but which policy is safe reverses. Source records cannot distinguish these targets without an additional target restriction. This construction establishes a scope limit; it does not assert that either target occurs in the experiments.

This section gives the complete conventional proof chain used by the paper. It includes the two short geometric arguments rather than leaving them as verbal sketches. Section EC.7 then maps the original proof chain to exact Lean declarations and separates proved implications from scientific premises.

EC.6.2. Profile-coordinate stability

Proof of Proposition 1. Let $e_d = h - \widehat{h}_d$. Since the integration interval has length one, the triangle inequality gives

$$\begin{aligned} |c_k(h) - c_k(\widehat{h}_d)| &= \left| \int_0^1 e_d(t) b_k(t) dt \right| \\ &\leq \int_0^1 |e_d(t)| |b_k(t)| dt \leq \epsilon_d B_k. \end{aligned}$$

For the regular-midpoint reconstruction, every t lies within $1/(2d)$ of the midpoint assigned to its Voronoi cell. Lipschitz continuity therefore implies

$$|h(t) - \widehat{h}_d(t)| \leq L_h/(2d),$$

so substituting $\epsilon_d = L_h/(2d)$ proves the first rate. Dividing both coefficients by $1 + \kappa k \geq 1$ divides their absolute difference by the same quantity and cannot enlarge it.

For completeness, suppose t lies between adjacent reference nodes l and r , has interpolation weight $\theta \in [0, 1]$, and the node spacing is $1/d_{\text{ref}}$. The interpolant is $\hat{h}(t) = (1 - \theta)h(l) + \theta h(r)$, whence

$$\begin{aligned} |h(t) - \hat{h}(t)| &\leq L_h \{(1 - \theta)|t - l| + \theta|t - r|\} \\ &= \frac{2L_h\theta(1 - \theta)}{d_{\text{ref}}} \leq \frac{L_h}{2d_{\text{ref}}}. \end{aligned}$$

This is the claimed pre-rounding interpolation bound.

EC.6.3. Farthest-first coverage

Proof of Theorem 1. Let $r = r(A_k; u)$ and choose a library point w whose distance to A_k is r . Farthest-first selection implies that the k selected centers together with w are pairwise separated by at least r : when each center was inserted, its distance from all earlier centers was at least the final covering radius, and the same is true of w by construction.

Assign these $k + 1$ witnesses to their nearest centers in an optimal k -center solution of radius r_k^* . Two witnesses, say x and y , must share one optimal center o . The triangle inequality then yields

$$r \leq \|u(x) - u(y)\|_2 \leq \|u(x) - u(o)\|_2 + \|u(o) - u(y)\|_2 \leq 2r_k^*.$$

The zero-radius case is immediate. Finally, deleting the two source-rank coordinates is a Euclidean projection and is nonexpansive. Hence an $\eta = (z, r^g, r^f)$ cover of radius r is also a z cover of radius at most r , proving Equation (14).

EC.6.4. Finite source errors and separated-rank recovery

Proof of Proposition 2 and the finite-source margin bound. Fix a source task and a library profile. For its R independent Gaussian constraint replications, write $\bar{Y}$ and S for the sample mean and sample standard deviation, and write μ and σ for their population counterparts. The Gaussian mean tail gives

$$\mathbb{P}\{|\bar{Y} - \mu| > r_\mu(\delta_\mu)\} \leq \delta_\mu.$$

Let $\nu = R - 1$. Cochran's identity gives $(R - 1)S^2/\sigma^2 \sim \chi_\nu^2$, and therefore, with probability at least $1 - \delta_\sigma$,

$$\sqrt{q_{\nu, \delta_\sigma/2}/\nu} \leq \frac{S}{\sigma} \leq \sqrt{q_{\nu, 1 - \delta_\sigma/2}/\nu}.$$

On this event,

$$|S - \sigma| \leq \sigma \max\{a_\nu(\delta_\sigma), b_\nu(\delta_\sigma)\} = r_\sigma(\delta_\sigma).$$

Because $\sigma \geq 10^{-3}$, replacing S by $\max\{S, 10^{-3}\}$ cannot increase its distance from σ . Hence, on the intersection of the mean and scale events,

$$\begin{aligned} |\hat{m} - m| &= |(\bar{Y} - \mu) + z_{1-\alpha}(\max\{S, 10^{-3}\} - \sigma)| \\ &\leq r_\mu(\delta_\mu) + z_{1-\alpha}r_\sigma(\delta_\sigma). \end{aligned}$$

The objective score uses the same argument without the scale term. Assigning the tail probabilities across the finite set of source tasks, profiles, and score heads and applying the union bound produces a simultaneous radius r_q for each score head.

It remains to show what this event implies for ranking. Let profiles j and ℓ have true scores $s_{qj} < s_{q\ell}$ with $s_{q\ell} - s_{qj} > 2r_q$. Uniform error at most r_q gives

$$\widehat{s}_{qj} \leq s_{qj} + r_q < s_{q\ell} - r_q \leq \widehat{s}_{q\ell}.$$

Thus every pair separated by more than $2r_q$ has the correct estimated order. Pairs inside that band may remain tied or reverse, which is why the result does not claim exact recovery of the full percentile ranking.

EC.6.5. From structural coverage to a feasible target policy

Proof of Theorem 2. By Assumption 1, choose a source-supported library profile h_s satisfying $d_q(h_s, h_q^\star) \leq \Delta_q$. Because the selected atlas covers the library in augmented coordinate η , there is an atlas member a such that

$$\|\eta(a) - \eta(h_s)\|_2 \leq r_{n_0}^\eta.$$

Projection onto the structural coordinates is nonexpansive, so

$$\|\widehat{z}(a) - \widehat{z}(h_s)\|_2 \leq r_{n_0}^\eta.$$

Insert $\widehat{z}(a)$, $\widehat{z}(h_s)$, and $z_q^\star(h_s)$ between the ideal target coordinates of a and the safe center. The triangle inequality and the two coordinate-approximation bounds give

$$\begin{aligned} d_q(a, h_q^\star) &\leq \|z_q^\star(a) - \widehat{z}(a)\|_2 + \|\widehat{z}(a) - \widehat{z}(h_s)\|_2 \\ &\quad + \|\widehat{z}(h_s) - z_q^\star(h_s)\|_2 + d_q(h_s, h_q^\star) \\ &\leq r_{n_0}^\eta + \Delta_q + 2\epsilon_\eta. \end{aligned}$$

The one-sided Lipschitz condition for the chance margin now yields

$$\begin{aligned} m_q(a) &\leq m_q(h_q^\star) + L_q d_q(a, h_q^\star) \\ &\leq -\gamma_q + L_q (r_{n_0}^\eta + \Delta_q + 2\epsilon_\eta) \leq 0, \end{aligned}$$

where the last inequality is exactly condition (20). Therefore a is target feasible and belongs to the selected atlas.

EC.6.6. Stratified task-law concentration

Proof of Equation (21). Index the independent held-out task outcomes by (s, i) and write $a_{si} = w_s/n_s$. Then

$$\hat{p} - p = \sum_s \sum_{i=1}^{n_s} a_{si} \{H_{si} - \mathbb{E}[H_{si}]\}.$$

The variables need not have the same distribution. Independence and $H_{si} \in [0, 1]$ are enough for Hoeffding's lemma, which gives

$$\mathbb{E} \exp\{\lambda(\hat{p} - p)\} \leq \exp\left\{\frac{\lambda^2}{8} \sum_s \sum_{i=1}^{n_s} a_{si}^2\right\} = \exp\{\lambda^2 v_w/8\},$$

because $\sum_i a_{si}^2 = w_s^2/n_s$. Chernoff optimization at $\lambda = 4r/v_w$ gives

$$\mathbb{P}\{\hat{p} - p \geq r\} \leq e^{-2r^2/v_w}.$$

Applying the same argument to $p - \hat{p}$ and taking a union bound yields $\mathbb{P}\{|\hat{p} - p| \geq r\} \leq 2e^{-2r^2/v_w}$. Set

$$r = \sqrt{\frac{v_w \log(2/\delta)}{2}}.$$

The right-hand side is δ , and on the complementary event $p \geq \hat{p} - r$, which is Equation (21). Applying the argument separately at each resolution is valid because independence is required within a resolution, not between the three reported resolutions.

EC.6.7. Exact familywise verification

Proof of Theorem 3. Consider a fixed candidate j whose true success probability satisfies $p_j < p_0$. Certification requires all v_j fresh Bernoulli trials to succeed. Independence within that candidate's verification stream gives

$$\mathbb{P}\{j \text{ certifies}\} = p_j^{v_j} \leq p_0^{v_j} \leq \delta_j.$$

If an unsafe policy is deployed from the fixed shortlist, at least one unsafe shortlist member must have certified. Consequently,

$$\mathbb{P}\{\text{an unsafe policy is deployed}\} \leq \sum_{j: p_j < p_0} \mathbb{P}\{j \text{ certifies}\} \leq \sum_{j=1}^{K_s} \delta_j \leq \delta.$$

This union bound does not require independence between candidates. Testing them in a fixed order and stopping at the first certificate only removes later tests from consideration; it does not enlarge the event that any unsafe candidate passes. Fixing the shortlist, its order, and all budgets before verification is what prevents search or selection from using the fresh test outcomes.

EC.6.8. Source-only scope, no-free-lunch, and call accounting

The frozen proposal is a function of source records and the declared target grid only. Replacing one unused target-label object by another therefore leaves the proposal unchanged. This is a definitional noninterference result, not a probabilistic generalization claim.

For the corresponding limitation, let $\mathcal{X}$ be finite and let a deterministic target-label-free atlas A be a proper subset of $\mathcal{X}$. Choose $x \in \mathcal{X} \setminus A$ and define the nonempty feasible set as $\mathcal{F} = \{x\}$. Then $A \cap \mathcal{F} = \emptyset$. In particular, $|A| \leq n_0 < |\mathcal{X}|$ implies that such a feasible set exists. Thus a finite source-only proposal needs the alignment premise in Theorem 2; coverage of every possible target is impossible.

Finally, if the source archive costs S simulations, target search costs N , and independent verification costs V , disjoint accounting gives $C_{\text{all}} = S + N + V$. If the same frozen archive is reused by M target problems, charging each target an equal share of the archive gives $C_{\text{amort}}(M) = S/M + N + V$. These are accounting identities; they do not claim that archive reuse is statistically valid for an unrelated target.

EC.7. Machine-Checked Proof Map

The Lean 4 project builds against mathlib without `sorry`, `admit`, or a project-defined `axiom`. The authoritative paper-facing manifest is `proof/SCOLHKG/PaperProofInterface.lean`. It contains compile-time checks for exactly the declarations in Table EC.14. Every original conventional proof in Section EC.6 maps to at least one declaration, and every declaration in the manifest maps back to a manuscript statement or accounting identity. Other files in the larger Lean archive formalize historical KG, HVD, and intermediate algorithm versions; they are not claims of this paper.

Proposition 3 is established by the analytic proof above and is outside the original Lean interface.

The map distinguishes logical closure from empirical truth. Assumption 1 is intentionally a premise consumed by the aligned-coverage theorem, not a theorem about every target. For replicated Gaussian source scores, the normal mean law and Cochran's chi-square identity are standard distributional inputs; Lean verifies the finite-sample variance identity, event union, variance-floor propagation, and rank implication after those inputs are supplied. Likewise, the task-law result assumes independent bounded held-out tasks, and terminal verification assumes the declared binomial law for each fresh stream. These boundaries are part of the theorem statements rather than unrecorded gaps.

EC.8. Reproducibility Materials

The original frozen evidence registry links 17,360 result cells to 14 experimental matrices and 12 compact analyses. It retains the one recorded algorithmic failure. Every row identifies the task, source archive, initial design, method, implementation version, verifier, source/search/verification calls, and the truth used only after the decision. Supplementary matched controls are recorded separately from these historical totals.

Table EC.14 Bidirectional closure of manuscript statements, conventional proofs, and the final paper-facing Lean interface.

Names after the file colon are exact Lean declarations.

Manuscript statement	Conventional proof	Lean file and declaration
Target-label noninterference	Section EC.6.8	<code>Real/RiskAlignedRepresentation.lean:</code> <code>sourceOnlyProposal_targetLabel_invariant</code>
Proposition 1 and interpolation rate	Section EC.6.2	<code>Real/ProfileCoordinateConsistency.lean:</code> <code>continuousProfileCoefficient_error_le;</code> <code>lipschitz_voronoi_coefficient_inverse_grid_rate;</code> <code>lipschitz_linear_interpolation_inverse_grid_rate;</code> <code>frequency_penalty_cannot_increase_coefficient_error</code>
Theorem 1 and projected cover	Section EC.6.3	<code>Real/FarthestFirstKCenter.lean:</code> <code>GonzalezWitnessCertificate.two_approx_every_k_center;</code> <code>Real/GeometricAtlasCoverage.lean:</code> <code>projected_coordinate_atlas_covers</code>
Proposition 2 and finite margin	Section EC.6.4	<code>Measure/SourceRankRecovery.lean:</code> <code>finite_source_profile_mean_bad_event_le_sum;</code> <code>floored_margin_bad_event_measure_le;</code> <code>Real/SourceRankRecovery.lean:</code> <code>floored_empirical_chance_margin_error_le;</code> <code>separated_source_pair_order_recovered</code>
Theorem 2	Section EC.6.5	<code>Real/GeometricAtlasCoverage.lean:</code> <code>finite_projected_aligned_geometric_lipschitz_atlas_coverage</code>
Equation (21)	Section EC.6.6	<code>Measure/TaskAtlasCoverage.lean:</code> <code>weightedTaskCoverageError_subGaussian;</code> <code>weightedTaskCoverageError_abs_tail_le</code>
Theorem 3	Section EC.6.7	<code>Measure/ExactBinomialCertificate.lean:</code> <code>unsafe_all_success_certificate_probability_le;</code> <code>finite_all_success_certificates_familywise</code>
Finite-atlas limitation	Section EC.6.8	<code>Real/ProposalNoFreeLunch.lean:</code> <code>proper_finite_atlas_misses_some_nonempty_feasible_set;</code> <code>finite_budget_no_unconditional_target_coverage</code>
Unamortized and amortized call identities	Section EC.6.8	<code>Real/MethodIndependentTerminalVerification.lean:</code> <code>paper_grade_budget_exact_decomposition;</code> <code>paper_grade_budget_amortized_decomposition</code>

The anonymous attachment includes compact JSONL projections of these cells, retaining identifiers, scalar outcomes, verifier results, costs, and the recorded failure. The September 7 comparisons add 960 synthetic and 180 Energy cells, with the standardized $d = 1000$ diagnostic reused among the synthetic cells. Their paired input records and analysis rules are included separately. The renderer reads compact JSON summaries to generate the empirical tables and figures. The package also contains the processed OPSD archive, current method and protocol files with the metadata erratum, and Lean source and dependency configuration. The September 30 evidence separately contains 4160 profile analysis rows (2400 attribution, 1440 prospective confirmation, and 320 library-size rows) and 720 corrected Energy V5 runs. The profile study executes 2360 new arm-task evaluations and reuses 1800 outcomes only for identical ordered designs; its confirmation has 240 distinct latent tasks. The HVD table renderer uses six aggregate input rows, rather than seed-level HVD records. Original geometric diagnostics retain their original scope; the displayed temporal diagnostics use V5.

EC.9. Public-battery development replay

This diagnostic uses the original 154 joint policy profiles, of which 100 are constant pairs, and three target seasons beginning April 15, July 15, and October 15, 2024. Each season has 73 supported window starts,

from 0 to 4320 at stride 60. The original controller and failure rule are fixed. The source archive contains 44,968 evaluated cells from two 2023 seasons, 154 profiles, 73 windows and two failure rules. The rules give identical outcomes and successful costs here. The source-selected center is `constant_pair_0007_0007`, with channel levels .75/.75.

Each search evaluates 13 distinct candidates on one uniformly sampled window per candidate. Ten initial policies are selected by farthest-first; three additional policies follow a common frozen random permutation. Successful samples rank by cost; failures keep null cost and rank after successes by evaluation order. A shortlist of three is then frozen. Verification uses an independent random stream to draw 80 window indices with replacement per candidate, stopping at the first all-success certificate. The per-candidate error allocation remains .05/3 and the reliability threshold remains .95. The source-singleton control searches only the source center and retains this conservative allocation.

The constant-only control selects its medoid and farthest-first points using standardized ordered-channel coordinates on the 100 constants, without target outcomes. It searches the same 13-candidate budget. Five seeds per season are algorithm repeats; they are averaged within season for cost accounting, not counted as separate deployments.

Design	Certificates/15	Target queries	Source queries
Source-first structural	15	279	44,968
Full-library structural	9	471	0
Constant-only structural	15	279	0
Source singleton	15	243	44,968

Target queries are the sum of within-season averages over three tasks. All four arms have zero false certificates relative to the fixed population. The new controls reuse stored cells and add no controller simulations; represented queries are charged even when a window is sampled repeatedly.

Paid census evaluates every supported window for every candidate. Its full-library query charge is $3 \times 154 \times 73 = 33,726$; the constant-only charge is $3 \times 100 \times 73 = 21,900$. Both select `constant_pair_0008_0009` in spring and `constant_pair_0006_0009` in summer and autumn. Their mean full-window costs are respectively £21,310.11, £44,648.60 and £33,568.78, with 73/73 successful windows each. The census ranks only candidates having a complete cost on every window; it is an exact paid finite-population calculation, not a free oracle.

Scope and limitations. The controls are post hoc development diagnostics, not independent confirmation. Window-index sampling supports inference only under the specified finite uniform law; overlapping historical windows do not establish calendar generalization. This is a complete-cost first-center adaptation, not the full rank-augmented atlas or the primary continuous search protocol. No 2025 observations are accessed. Code, protocols and compact summaries are retained in the project under the `public_battery_simple_controls_v1_20261002` identifier in the attachment specified in Section 10 of the main paper. Window-level replay requires the retained source and target population journals, which are outside the compact attachment.

EC.10. First-center margin diagnostic

After all confirmation decisions were fixed, we reconstructed the same 24 source archives using their original seeds and replayed 9,216 source replications. Every estimated first center matched its stored decision. No new independent source samples or target simulations were collected. For the original 240 confirmation tasks we evaluated generator truth on all 64 mapped library policies (15,360 deterministic evaluations). Population objective and chance-margin ranks define b_j with the original equal weights and tie breaks. The diagnostic radius is the observed $\max_j |\widehat{b}_j - b_j|$, not a confidence radius.

For this finite-domain check choose $z_q^* = \widehat{z}$, so coordinate error is zero, and compute the exact maximum of $|m_q(h_i) - m_q(h_j)| / \|\widehat{z}(h_i) - \widehat{z}(h_j)\|_2$ over distinct coordinates in the mapped library. No coordinate collision with unequal margins occurred. This supplies a Lipschitz constant only on that finite library, not on the continuous policy space.

The sufficient condition of Proposition EC.1 holds on 0/240 tasks. Source-compatible sets have 14–33 policies. Population-score first centers are target feasible in 119/240 tasks; estimated first centers are feasible in 106/240. These totals do not imply that removing source noise improves every task: for example, piecewise smoothness has 14/30 and 22/30 respectively. The proposed bound is therefore empirically vacuous in this diagnostic. It remains a conditional analytic statement, not an established explanation of the experimental gain or a selection rule. Compact results and code are retained under `first_center_margin_diagnostic_20261002` in the current submission attachment.

EC.11. Fixed-first-center remainder controls

These rules were fixed after observing the 240-task confirmation and its mechanism diagnostics. The comparisons are postdecision evidence on that same task population. For each original task the random control retains the source-selected first policy and draws nine distinct candidates uniformly from the other 63 IDs using `SeedSequence(family_seed, design_seed, 53107)`. The entire original library remains eligible, without target-outcome filters. The other new control initializes the original source-greedy portfolio with this same first policy, then applies its original score and tie rules to nine further choices. It reuses 230 results only where all ten ordered IDs match; ten changed ordered designs are evaluated anew. The random control adds 240 arm-task runs. All use $d = 1000$, $N = n_0 = 10$ and the unchanged (80, 80, 80) all-success verifier at familywise error .05.

Design	Design hits	Shortlist hits	True certificates	False
Source-first + random nine	167	158	53	0
Source-first + structural	155	155	54	1
Source-scored atlas	155	155	56	1
Fixed-first portfolio	212	177	49	1

All denominators are 240 tasks. Random-remainder mean penalized search-set loss is .361, versus .414 for structural coverage and .156 for the fixed-first portfolio. This hidden-truth search-set diagnostic is distinct

from certified deployment cost. The atlas-minus-random certification interval is $[-2.5, 5.0]$ percentage points; structural-minus-random is $[-3.3, 4.2]$. Neither establishes superiority or equivalence.

All 24 original archives are replayed during preparation using their original source streams, adding no independent source evidence. Represented source cost is 9,216 calls paid once per source-based arm across the 240 tasks. The 250 new runs consume 2,500 target-search and 52,160 verification calls. Cumulative source-inclusive calls are 62,576 for random remainder, 60,976 for structural coverage, 60,736 for the atlas and 63,936 for the fixed-first portfolio. Differences in certification remain unresolved.

For a frozen shortlist with candidate feasibility probabilities p_j , let $q_j = p_j^{80}$. Under the declared independent simulation streams, the conditional probability of a true certificate is

$$\sum_{j: p_j \geq .95} q_j \prod_{\ell < j} (1 - q_\ell).$$

Averaging this expression over the fixed task shortlists gives expected true-certificate counts 48.60, 49.44, 49.72 and 54.07 in the table's order. These quantities use generator truth for explanation; they do not supply new certificates or average over new search observations.

Scope and limitations. Only one random remainder is frozen per original design seed; the tasks, source archives and libraries have already been observed. The controls do not authorize tuning the library, ranks, shortlist or verifier. The .99 risk-probability summaries in the compact artifacts are descriptive only; the chance threshold remains .95. No 2025 data are accessed. Code and compact artifacts are included in the current submission attachment under `source_first_remainders_v1_20261002`.

References

Feurer M, Eggenberger K, Falkner S, Lindauer M, Hutter F (2022) Auto-Sklearn 2.0: Hands-free AutoML via meta-learning. *Journal of Machine Learning Research* 23(261):1–61, URL <https://jmlr.org/papers/v23/21-0992.html>.